AI 500 - Practice Problem Set 4

1. MCQs:

Decision Trees (Entropy and Information Gain)

Q1.

Consider a dataset with 12 instances: 8 labeled as "Buy" and 4 labeled as "Don't Buy". What is the entropy of this dataset?

A. $E = -(8/12)\log_2(8/12) - (4/12)\log_2(4/12) \approx 0.918$
B. $E = -(8/12)\log_2(8/12) + (4/12)\log_2(4/12) \approx -0.918$
C. $E = (8/12)\log_2(8/12) + (4/12)\log_2(4/12) \approx -0.918$
D. E = 0 because there are only two classes

Q2.

A leaf node in a decision tree is created when certain stopping criteria are met. Which of the following are valid stopping criteria?

A. All instances at the node belong to the same class (n_labels == 1)
B. The maximum depth of the tree has been reached
C. The number of samples at the node falls below min_samples_split
D. The information gain for all possible splits is positive

Q3.

Consider splitting a node with entropy E = 1.0 (perfect uncertainty). After evaluating all features, the best split achieves an information gain of 0.03. What does this indicate?

A. The split provides very little improvement in class purity
B. This split should be avoided, and the node should become a leaf
C. A low information gain near 0 suggests that the feature does not effectively separate the classes
D. The weighted average entropy of the child nodes is approximately 0.97

Random Forests

Q4.

In a Random Forest with 100 decision trees, how is the final prediction made for a classification task?

A. The prediction from the tree with the highest accuracy on validation data is selected
B. Majority voting: the class predicted by most trees becomes the final prediction
C. The predictions are averaged, and the class closest to the mean is selected
D. Each tree's prediction is weighted by its depth, and the weighted majority is taken

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Q5.

The Random Forest training process involves bootstrap sampling. Which statements correctly describe this process?

A. Bootstrap sampling creates multiple subsets by randomly selecting instances with replacement from the original dataset
B. Each subset has the same size as the original training dataset
C. Some instances may appear multiple times in a single bootstrap sample, while others may not appear at all
D. Bootstrap sampling guarantees that each tree sees exactly the same training data

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Q6.

What are the key advantages of Random Forests compared to single decision trees?

A. Random Forests are less prone to overfitting because they average predictions across multiple trees
B. Random Forests always have lower bias than single decision trees
C. Random Forests can capture more complex decision boundaries by combining multiple trees
D. Random Forests require less training time than a single decision tree

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Overfitting/Underfitting + Bias/Variance

Q7.

A decision tree model achieves 98% accuracy on the training set but only 65% accuracy on the validation set. What problem does this indicate?

A. The model is overfitting the training data
B. The model has high variance and captures noise in the training data
C. The model should be simplified by reducing its depth or increasing min_samples_split
D. This is expected behavior and indicates a well-fitted model

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Q8.

In the bias-variance tradeoff for decision trees, which statements are correct?

A. Deep trees typically have low bias but high variance
B. Shallow trees typically have high bias but low variance
C. Increasing tree depth generally reduces bias but increases variance
D. Random Forests address high variance by averaging predictions across multiple trees

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Q9.

You observe that your decision tree has 100% accuracy on training data but 60% on validation data. Which techniques could help address this issue?

A. Increase the maximum depth of the tree to capture more patterns
B. Increase min_samples_split to prevent the tree from creating very specific leaf nodes
C. Use ensemble methods like Random Forests or Bagging
D. Collect more training data to better represent the underlying distribution

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Bagging (Bootstrap Aggregation)

Q10.

In Bagging, multiple base models are trained. Which statements correctly describe the Bagging process?

A. Base models are trained independently and in parallel on different subsets of the training data
B. Each subset is generated using bootstrap sampling (random sampling with replacement)
C. For classification, the final prediction is made by majority voting across all base models
D. Bagging primarily reduces bias in the models

Q11.

Consider a bagging ensemble with 10 decision trees for binary classification. For a test instance, 7 trees predict "Yes" and 3 predict "No". What is the final prediction?

A. "Yes" because it received the majority vote (7 > 3)
B. "No" because we should be conservative and predict the minority class
C. For regression tasks, Bagging would average the predictions instead of voting
D. The confidence in the "Yes" prediction can be estimated as 70%

Q12.

What is the primary goal of using Bagging in ensemble learning?

A. To reduce variance and prevent overfitting
B. To reduce bias and improve model accuracy
C. To make the model more robust by combining predictions from multiple diverse models
D. To increase training speed by parallelizing model training

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Boosting

Q13.

How does Boosting differ fundamentally from Bagging in terms of model training?

A. In Boosting, models are built sequentially, with each model trying to correct errors of previous models
B. In Boosting, all models are trained independently and in parallel
C. Boosting focuses on reducing bias rather than variance
D. In Boosting, models are weighted according to their performance, unlike Bagging where all models have equal weight

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Q14.

Consider a Boosting algorithm where Model 1 has high error on certain instances. What happens in the training of Model 2?

A. Model 2 gives higher weight to the instances that Model 1 misclassified
B. Model 2 focuses on correcting the errors made by Model 1
C. Model 2 is trained on a completely different bootstrap sample unrelated to Model 1's performance
D. The sequential nature of Boosting means each model is influenced by previous models' performance

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Q15.

In Boosting, why are models weighted according to their performance during final prediction?

A. Models with better accuracy should have more influence on the final prediction
B. All models should have equal weight to ensure fairness
C. Weighting allows the ensemble to emphasize strong learners and de-emphasize weak learners
D. The weighting scheme is a key difference from Bagging, where all models have equal weight

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Isolation Forests

Q16.

Isolation Forest is designed for anomaly detection. What are the key assumptions that Isolation Forest relies on?

A. Anomalies are the minority, consisting of fewer instances
B. Anomalies have attribute values that are very different from normal instances
C. Anomalies always form tight clusters separate from normal data
D. Normal instances require more partitions (deeper paths) to isolate than anomalies

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Q17.

The anomaly score in Isolation Forest is calculated as $s(x, n) = 2^{(-E(h(x))/c(n))}$, where $E(h(x))$ is the average path length and $c(n)$ is used for normalization. Which statements about interpreting the anomaly score are correct?

A. If $s \approx 1$, the instance is definitely an anomaly
B. If $s << 0.5$ (much smaller than 0.5), the instance is likely a normal instance
C. If all instances have $s \approx 0.5$, the sample does not contain distinct anomalies
D. The anomaly score ranges from 0 to 1

Q18.

During the training stage of Isolation Forest, multiple isolation trees (iTrees) are built using sub-sampled data. During the evaluation stage, a test instance is passed through all trees to compute the anomaly score. If an instance has $E(h(x)) \rightarrow c(n)$, what does this indicate?

A. The instance's average path length is similar to the expected path length for normal instances
B. The anomaly score $s(x, n) \rightarrow 0.5$, suggesting the instance is neither clearly anomalous nor clearly normal
C. This instance should be classified as an anomaly
D. The instance is difficult to isolate, behaving similarly to the bulk of normal instances

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SVM

Question 19

Which of the following statements best describes the role of Support Vectors in a trained linear SVM model?

A) They are the outliers that are ignored by the algorithm to create a more robust decision boundary.

B) They are all the data points that are correctly classified by the hyperplane.

C) They are the specific data points that lie on the margin; removing any other point from the dataset would not change the decision boundary.

D) They are used to calculate the mean for the least-squares error solution before finding the hyperplane.

Question 20

What is the primary advantage of using the Kernel Trick in SVMs for non-linearly separable data?

A) It simplifies the model by reducing the number of features in the original dataset, making the computation faster.

B) It explicitly transforms the data into a higher-dimensional space, and then uses linear regression to find the separating plane.

C) It allows the SVM to learn a complex, non-linear decision boundary by computing the dot product in a higher-dimensional space efficiently, without ever explicitly creating the high-dimensional data points.

D) It is a regularization technique used to automatically find the optimal value for the parameter λ (lambda) to prevent overfitting.

2. Numerical Questions:

Entropy, Gini, Information Gain

Q19.

You have 8 samples with one numeric feature x and label y $\in \{0, 1\}$

x	2	3	4	5	6	7	8	9
y	0	0	0	1	1	1	1	1

A. Calculate the entropy of the parent.

B. Info gain if you split at $x \leq 4.5$?

C. Info gain if you split at $x \leq 6.5$?

D. Which split is better?

Q20.

A node has 10 samples: 6 positives and 4 negatives. Candidate split:

- Left child: 5 samples (4 pos, 1 neg)
- Right child: 5 samples (2 pos, 3 neg)

A. Compute parent Gini.

B. Compute child Ginis.

C. Compute the weighted Gini of the child nodes.

D. What is the Gini gain from the split?

SVM

You are given a 2D dataset with three data points:

- **Point 1 (Class +1):** P1 = (3, 4)
- **Point 2 (Class -1):** P2 = (5, 2)
- **Point 3 (Class +1):** P3 = (2, 5)

Instructions:

1. Identify which two points are the **support vectors**. (Hint: For a linearly separable dataset, the support vectors are the pair of points from opposite classes with the smallest Euclidean distance between them).
2. Using the support vectors you identified, calculate the equation of the decision boundary in the form $w^Tx - b = 0$.
3. Calculate the width of the margin.

Q. Linear Regression — Least Squares Fit

A dataset contains the following pairs of (x, y):

X	Y
1	1
2	2
3	1.3
4	3.75
5	2.25

Tasks:

A. Compute the least squares estimates of slope b and intercept a for the regression line y=a+bx

B. Write the equation of the fitted line.

C. Predict the value of y when x=6x
D. Compute the sum of squared errors (SSE) for the fitted model.

Q. SVM with Polynomial Kernel

Consider a simple 2D dataset:

Point	(x_1, x_2)	Class
P_1	(1, 1)	+1
P_2	(-1,-1)	+1
P_3	(1, -1)	-1
P_4	(-1, 1)	-1

You will use a polynomial kernel K(x,z)=(x · z)^2.

Tasks:
A. Compute the kernel matrix Kij for all four points.
B. Using symmetry, determine the Lagrange multipliers αi assuming all support vectors contribute equally.
C. Compute the bias term b.
D. Write the final decision function f(x)=∑αi yi K(xi,x)+b.
E. Predict the class for the test point x=(0.5,0.5).

Decision Trees (Entropy & Information Gain)

Problem 1: Student Performance Prediction

A dataset contains 40 students classified by whether they passed (P) or failed (F):

- Overall: 28 passed, 12 failed

Two features are being considered for the first split:

Feature A - Study Hours:

- Low (0-2 hrs): 5P, 8F
- Medium (3-5 hrs): 10P, 3F
- High (6+ hrs): 13P, 1F

Feature B - Attendance:

- Poor (<60%): 3P, 7F
- Good (60-85%): 12P, 4F
- Excellent (>85%): 13P, 1F

Calculate:

1. Information gain for Feature A
2. Information gain for Feature B
3. Which feature should be selected as the root node split?
4. After selecting the best feature, what would be the entropy of the node with the highest remaining impurity?

Random Forests

Problem 1: Bootstrap Sampling Analysis

You're training a Random Forest with 3 trees using a dataset of 10 instances: {A, B, C, D, E, F, G, H, I, J}

Bootstrap samples created:

- Tree 1: {A, A, C, D, E, E, E, G, I, J}
- Tree 2: {B, C, C, D, F, F, G, H, I, J}
- Tree 3: {A, B, D, D, E, F, G, G, H, I}

For a test instance X, the three trees predict:

- Tree 1: Class "Positive"
- Tree 2: Class "Negative"
- Tree 3: Class "Positive"

Answer the following:

1. Which instances from the original dataset were never selected in Tree 1's bootstrap sample?
2. What is the final prediction for instance X?
3. What confidence level (as a percentage) can you assign to this prediction?

Problem 2: Forest Prediction Aggregation

A Random Forest classifier with 7 trees is used for disease diagnosis (Disease/Healthy). For patient P, individual tree predictions are:

Trees 1-7 predict: Disease, Healthy, Disease, Disease, Healthy, Disease, Disease

Calculate:

1. The final classification for patient P
2. The prediction confidence percentage
3. If you needed 70% confidence to recommend treatment, would you proceed? Why?
4. How many trees would need to change their prediction to reverse the final classification?

Overfitting/Underfitting + Bias/Variance

Problem 1: Model Performance Comparison

Three decision tree models with different depths are trained and evaluated:

Model A (Depth = 3):

- Training accuracy: 73%
- Validation accuracy: 71%

Model B (Depth = 8):

- Training accuracy: 89%
- Validation accuracy: 86%

Model C (Depth = 15):

- Training accuracy: 99%
- Validation accuracy: 68%

Answer the follwoing:

1. Calculate the training-validation gap for each model
2. Which model shows signs of overfitting? Provide evidence.
3. Which model would you deploy and why?
4. Which model likely has the highest variance?

Problem 2: Performance Degradation Analysis

You train a decision tree on 1000 samples and observe the following:

Training Progress:

- At depth 5: Train accuracy = 82%, Validation accuracy = 80%
- At depth 10: Train accuracy = 94%, Validation accuracy = 83%
- At depth 15: Train accuracy = 99%, Validation accuracy = 77%

Calculate:

1. The performance degradation on validation set from depth 10 to depth 15
2. At which depth does overfitting start to become problematic?
3. If you had to choose between depth 10 and depth 5, which would you pick and why?

Isolation Forests

Problem 1: Path Length Calculation

An Isolation Forest with 3 trees evaluates instance X. The path lengths from root to leaf for X in each tree are:

- Tree 1: 4 edges
- Tree 2: 3 edges
- Tree 3: 5 edges

The dataset size n = 100, so c(n) ≈ 10.29 (given for normalization).

Calculate:

1. E(h(X)) - the average path length for instance X
2. The anomaly score s(X, n) = 2^(-E(h(X))/c(n))
3. Is instance X likely an anomaly? (Use threshold: s > 0.6 suggests anomaly)

Problem 2: Isolation Forest Evaluation

An Isolation Forest with 4 trees evaluates three instances. Path lengths (in edges) are:

Instance	Tree 1	Tree 2	Tree 3	Tree 4
P	8	7	9	8
Q	3	2	4	3
R	6	5	7	6

Dataset size n = 200, therefore c(n) ≈ 10.66.

Calculate:

1. E(h(X)) for each instance (P, Q, R)
2. Anomaly scores for all three instances
3. Rank the instances from most to least anomalous
4. If you need to flag the top 30% as anomalies for investigation, which instance(s) would you flag?
5. What does the relationship between path length and anomaly score tell you about how Isolation Forest identifies outliers?